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①. Gen. sol?

a) $x_{k+2} - \sqrt{3}x_{k+1} + x_k = 0$ second order LTDE w/ c.c.

$$r^2 - \sqrt{3}r + 1 = 0$$

$$\Delta = -1$$

$$r_{1,2} = \frac{\sqrt{3} \pm i}{2}$$

$$\rightarrow \operatorname{Re}\left(\frac{\sqrt{3}}{2} + \frac{i}{2}\right)^k, \operatorname{Im}\left(\frac{\sqrt{3}}{2} + \frac{i}{2}\right)^k$$

$$z = \frac{\sqrt{3}}{2} + \frac{i}{2} = \rho(\cos\theta + i\sin\theta) = 1 \cdot \left(\cos\frac{\pi}{6} + i\sin\frac{\pi}{6}\right).$$

$$\rho = \sqrt{\left(\frac{\sqrt{3}}{2}\right)^2 + \left(\frac{1}{2}\right)^2} = \sqrt{\frac{3}{4} + \frac{1}{4}} = \sqrt{\frac{4}{4}} = 1.$$

$$\cos\theta = \frac{\sqrt{3}}{2 \cdot 1} > 0 \Rightarrow \text{cI}, \theta = \frac{\pi}{6}$$

$$\sin\theta = \frac{1}{2 \cdot 1} > 0$$

$$x = c_1 \cdot \cos\frac{k\pi}{6} + c_2 \cdot \cos\frac{k\pi}{6}$$

b) $x'' - \sqrt{3}x' + x = 0.$

$$r^2 - \sqrt{3}r + 1 = 0$$

$$r_{1,2} = \frac{\sqrt{3}}{2} \pm \frac{i}{2} \rightarrow e^{\frac{\sqrt{3}}{2}t} \cdot \cos\frac{1}{2}t, e^{\frac{\sqrt{3}}{2}t} \cdot \sin\frac{1}{2}t$$

$$\Rightarrow x = e^{\frac{\sqrt{3}}{2}t} \cdot \cos\frac{t}{2} + c_2 \cdot e^{\frac{\sqrt{3}}{2}t} \cdot \sin\frac{t}{2}$$

$$\textcircled{2} \begin{cases} \dot{x} = y \\ \dot{y} = -g \sin x \end{cases}$$

$$a) \frac{dy}{dx} = \frac{\dot{y}}{\dot{x}} \quad (2) \quad \frac{-g \sin x}{y} = \frac{dy}{dx} \Rightarrow -g \sin x dx = y dy \quad | \int \Rightarrow$$

$$\Rightarrow -g \int \sin x dx = \int y dy \Rightarrow -g \cdot (-\cos x) = \frac{y^2}{2} + c \Rightarrow$$

$$\Rightarrow \frac{y^2}{2} - g \cos x = c$$

$$\Rightarrow H: \mathbb{R}^2 \rightarrow \mathbb{R}, H(x, y) = \frac{y^2}{2} - g \cos x.$$

$$H \text{ f.i. (global)} \Leftrightarrow \frac{\partial H}{\partial x}(x, y) \cdot (\dot{x}) + \frac{\partial H}{\partial y}(x, y) \cdot (\dot{y}) = 0$$

$$\frac{\partial H}{\partial x} = g \sin x, \frac{\partial H}{\partial y} = y.$$

$$\Rightarrow (g \sin x) \cdot y + y \cdot (-g \sin x) = 0, \text{ true} \Rightarrow H \text{ f.i.}$$

b) $(2\pi, 0)$ equil. point? hyp? stable?

$$A = \begin{pmatrix} 0 & y \\ -g \sin x & 0 \end{pmatrix}$$

$(2\pi, 0)$ equil. point because $\begin{cases} \dot{x} = y = 0 \\ \dot{y} = -g \sin x = -g \cdot \sin(2\pi) = 0. \end{cases}$

$$Jf(x, y) = \begin{pmatrix} 0 & 1 \\ -g \cos x & 0 \end{pmatrix}$$

$$Jf(2\pi, 0) = \begin{pmatrix} 0 & 1 \\ -g \cdot 1 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ -g & 0 \end{pmatrix}$$

$$\begin{vmatrix} -\lambda & 1 \\ -g & -\lambda \end{vmatrix} = 0 \Rightarrow \lambda^2 + g = 0 \Rightarrow \lambda = \pm \sqrt{-g} \Rightarrow \text{center} \Rightarrow \text{stable.}$$

$\text{Re}(\lambda) = 0 \Rightarrow \text{not hyperbolic}$

$$H(x, y) = \frac{y^2}{2} - g \cos x$$

$$H(2\pi, 0) = 0 - g \cdot \cos(2\pi) = -g.$$

For the ideal pendulum, linearization method fails at center-type eq., so use the first integral. Around $(2\pi, 0)$, the orbits are closed curves encircling the equilibrium $\Rightarrow$ stable eq.

③. $h > 0$? s.t. attractor of $x' = x^2 + 5x + 6$ is attractor also for the discrete dyn. sys. associated to Euler's num. formula with $h > 0$.

$$x' = x^2 + 5x + 6 = 0 \Rightarrow \eta_1^* = -3$$

$$\text{let } f(x) = x^2 + 5x + 6 \quad \eta_2^* = -2$$

$$\text{dim. method: } f'(x) = 2x + 5$$

$$f'(-3) = -1 < 0 \Rightarrow \underline{-3 \text{ attractor}}$$

$$f'(-2) = 1 > 0 \Rightarrow \text{repeller.}$$

Euler: $x_k = k \cdot h$
 ~~$y_{k+1} = y_k + h \cdot y_k$~~

$$\boxed{g(y) = y + h \cdot f(y)}$$

$$g(y) = y + h \cdot (y^2 + 5y + 6)$$

-3 attractor for the discrete dyn. system $\Leftrightarrow |g'(-3)| < 1$

$$g'(y) = 1 + h \cdot (2y + 5)$$

$$|g'(-3)| = |1 + h \cdot (-4)| = |1 - h| < 1 \Rightarrow -1 < 1 - h < 1 \quad | \cdot (-1)$$

$$\Rightarrow -2 < -h < 0 \quad | \cdot (-1)$$

$$\Rightarrow 0 < h < 2$$